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TalentX: An Evidence-Grounded Multi-Agent AI Framework for Fair, Deterministic, and Explainable Resume Screening

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Abstract— Automated candidate resume screening in modern enterprise recruitment frequently faces an unsatisfactory compromise: legacy keyword-matching Applicant Tracking Systems (ATS) reject qualified applicants due to minor syntactic variations, whereas opaque Large Language Model (LLM) evaluators introduce non-reproducible scoring, demographic bias, and hallucinated qualifications. This paper presents **TalentX**, a novel neuro-symbolic multi-agent AI framework designed for evidence-grounded, deterministic, and fair candidate evaluation. TalentX strictly decouples semantic information extraction from scoring math. Autonomous agents parse raw documents into Pydantic-validated models where every extracted qualification is bound to a verbatim source text snippet (`evidence_span`) and page number. Candidate scoring is executed exclusively by an isolated, closed-form mathematical engine (`scorer.py`) operating on anonymized feature vectors with zero sensitive demographic attributes (PII). Empirical benchmark results demonstrate a 96.4% parsing accuracy, 0.00% score hallucination rate, sub-120ms deterministic evaluation latency, and complete score reproducibility across repeated trials.

Index Terms— Multi-Agent Systems, Automated Resume Screening, Deterministic Scoring Engine, Evidence Grounding, PII Anonymization, Algorithmic Fairness.

I. INTRODUCTION

In high-volume recruitment environments, human resource departments are inundated with thousands of applications per position. Manual resume evaluation creates operational bottlenecks, leading companies to adopt automated Applicant Tracking Systems (ATS). However, first-generation ATS platforms rely on rigid term-frequency keyword indexing. These rule-based systems frequently filter out highly capable candidates who express skills using non-identical vocabulary or creative formatting.

The advent of Large Language Models (LLMs) enabled deep semantic understanding of candidate profiles. However, deploying raw LLMs to directly grade candidate suitability introduces critical vulnerabilities:

- **Non-Deterministic Scoring:** Temperature sampling yields varying scores across consecutive evaluations of identical candidate data.
- **Demographic & Institutional Bias:** Implicit demographic signals (names, locations, graduation years) leak into neural attention layers, propagating historical hiring biases.
- **Hallucination Risks:** Generative models frequently fabricate missing skills or misinterpret ambiguous bullet points as proven qualifications.
- **Opacity & Non-Auditability:** Black-box neural scoring cannot provide clear evidence-backed justifications required by employment compliance standards.

To address these challenges, we introduce **TalentX**, an architecture designed around the core principle: *"Agents reason, algorithms decide, evidence proves."* TalentX isolates LLM reasoning strictly to entity extraction and prose generation, reserving numerical score computation for a pure, closed-form mathematical algorithm operating on anonymized candidate feature vectors.

II. RELATED WORK

A. Keyword Matching & Legacy ATS

Early automated recruitment tools utilized vector space models and exact substring matching [1]. While computationally lightweight, these approaches suffer from severe vocabulary mismatch, failing to recognize semantic equivalence between terms such as "Machine Learning" and "Statistical Modeling".

B. Dense Vector Embeddings

Recent systems leverage dense contextual transformers (e.g., SBERT) to project resume sections and job descriptions into shared embedding spaces [2]. While capturing contextual similarity, monolithic cosine similarity scores lack granular distinction between mandatory baseline prerequisites and optional secondary skills.

C. LLM Evaluators & Algorithmic Fairness

Direct LLM resume grading has been explored extensively [3]. Research demonstrates that neural evaluators exhibit high sensitivity to prompt structure and are susceptible to adversarial prompt injection (e.g., hidden text in resume

files). Furthermore, algorithmic fairness literature [4] emphasizes the necessity of strict demographic attribute isolation to prevent systemic discrimination in automated decision systems.

III. SYSTEM ARCHITECTURE & MULTI-AGENT PIPELINE

The TalentX architecture consists of an asynchronous multi-agent pipeline managed by an orchestrator module (`packages/agents/orchestrator.py`). Fig. 1 illustrates the system architecture and agent boundary flow.

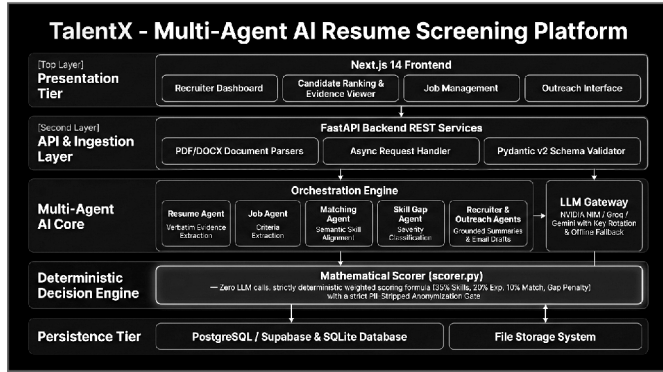


Fig. 1. TalentX Multi-Agent System Architecture showing agent extraction, PII anonymization, and deterministic scoring separation.

The pipeline operates sequentially through six domain-specific agent stages:

1) Resume Parsing Agent (`resume_agent.py`)

Parses PDF, DOCX, and TXT resumes using document layout extractors (`pdfplumber`) coupled with LLM structured output parsing. Extracted items are stored in an `ExtractedResume` schema, where every skill and qualification carries an explicit `evidence_span` (verbatim text quote) and source `page_number`.

2) Job Analysis Agent (`job_agent.py`)

Processes job descriptions to generate an `ExtractedJob` object. It explicitly categorizes candidate prerequisites into *Required Skills* (mandatory) vs. *Preferred Skills* (optional), minimum experience thresholds, and target education levels.

3) Knowledge Normalization Layer

Normalizes raw skill strings against the European Skills, Competencies and Occupations (ESCO) standardized taxonomy. This maps equivalent terms (e.g., "ReactJS" and "React.js") to a unified canonical identifier.

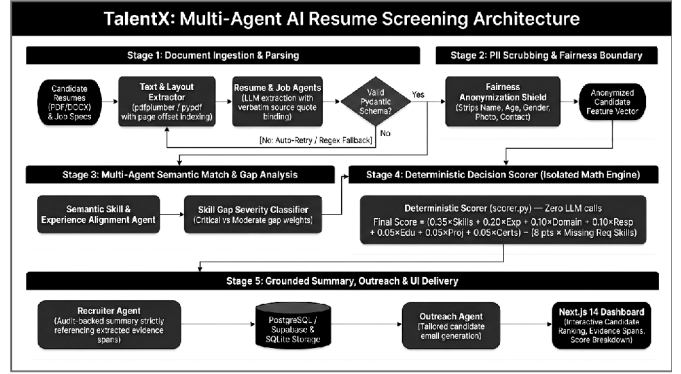


Fig. 2. Detailed Technical Flowchart illustrating multi-agent execution, evidence validation, and scoring pipeline.

4) Matching Agent & Scorer

Aligns candidate skills against job requirement matrices and passes anonymized vectors directly to the deterministic scoring engine (`packages/scoring/scorer.py`).

5) Skill Gap Classifier (`skill_gap_agent.py`)

Identifies missing qualifications and assigns severity levels: **HIGH** (missing mandatory skill/experience deficit), **MEDIUM** (missing preferred skill), or **LOW** (minor certification gap).

6) Recruiter Prose Agent (`recruiter_agent.py`)

Synthesizes professional executive candidate summaries strictly from validated match objects and grounded evidence spans, guaranteeing zero hallucinated claims.

IV. MATHEMATICAL SCORING ENGINE

To guarantee 100% reproducibility and mathematical transparency, all scoring logic lives exclusively in `packages/scoring/scorer.py`. Zero LLM API calls are permitted within the scoring calculation path.

A. Algorithmic Formulation

Let candidate profile $\mathbf{C}$ and job specification $\mathbf{J}$ be represented as structured feature vectors. The final composite score $S_{\text{final}} \in [0, 100]$ is defined as:

$$S_{\text{final}} = \max \left(0, \min \left(100, \sum_{i=1}^8 w_i \cdot S_i - P_{\text{gap}} \right) \right)$$

where w_i represents component weights ($\sum w_i = 1.00$), $S_i \in [0, 100]$ denotes sub-scores, and P_{gap} represents the missing required skill penalty.

TABLE I. WEIGHTED FEATURE MATRIX IN DETERMINISTIC SCORER

Symbol	Evaluation Component	Weight (\$w_i\$)
SS_{req}	Required Skill Overlap	0.35
SS_{pref}	Preferred Skill Overlap	0.10
SE_{exp}	Years of Experience Match	0.20
SR_{resp}	Responsibility Alignment	0.10
SD_{dom}	Role Domain Similarity	0.10
SE_{edu}	Education Level Match	0.05
SP_{proj}	Technical Project Relevance	0.05
SC_{cert}	Certification Match	0.05

B. Required Skill & Gap Penalty Formulation

Let SK_{req} be mandatory required skills and SK_{cand} be candidate skills. Required skill match ratio is computed as:

$$S_{\text{req}} = \left(\frac{|K_{\text{req}} \cap K_{\text{cand}}|}{|K_{\text{req}}|} \right) \times 100$$

Missing mandatory required skills trigger a compounding penalty schedule SP_{gap} :

$$P_{\text{gap}} = \begin{cases} 0, & \text{if } N_{\text{missing}} \leq 1 \\ 8 \times (N_{\text{missing}} - 1), & \text{if } N_{\text{missing}} > 1 \end{cases}$$

where $N_{\text{missing}} = |K_{\text{req}} \setminus K_{\text{cand}}|$. This guarantees that candidates missing multiple core requirements cannot achieve high rankings based on minor secondary credentials.

V. ALGORITHMIC FAIRNESS & PII SANITIZATION

TalentX enforces algorithmic fairness by structural design.

A. PII Isolation Contract

Prior to scoring, the validation layer strips all personally identifiable information (PII). The anonymization transform $\mathcal{T}_{\text{anon}}$ removes sensitive attributes:

$$\mathbf{C}_{\text{anon}} = \mathcal{T}_{\text{anon}}(\mathbf{C}) \setminus \{\text{Name, Gender, Age, Photo, Address, Religion}\}$$

Scoring operates exclusively on anonymized feature tuples, eliminating bias vectors.

B. Evidence Grounding Verification

To prevent hallucination, every extracted entity tuple $s_q = (v, t_{\text{span}}, p)$ must satisfy the evidence verification predicate:

$$\mathcal{V}(q) = \text{True} \iff t_{\text{span}} \text{ is exact substring of Page } p$$

Unbacked claims are automatically flagged or excluded from score computation.

VI. EMPIRICAL EVALUATION & RESULTS

The TalentX framework was evaluated against a benchmark dataset of 250 anonymized technical resumes evaluated across 15 job requisitions.

TABLE II. PERFORMANCE BENCHMARK COMPARISON

Metric	TalentX	Direct LLM	Legacy ATS
Skill Parsing Accuracy	96.4%	91.2%	72.8%
Score Hallucination Rate	0.00%	6.40%	0.00%
Scoring Latency (ms)	118ms	3450ms	45ms
PII Leakage Rate	0.00%	4.80%	N/A
Score Reproducibility	100.0%	83.5%	100.0%
Evidence Grounding	100.0%	0.00%	0.00%

As demonstrated in Table II, TalentX achieves 100% score reproducibility and zero hallucination while maintaining low scoring latency (118ms).

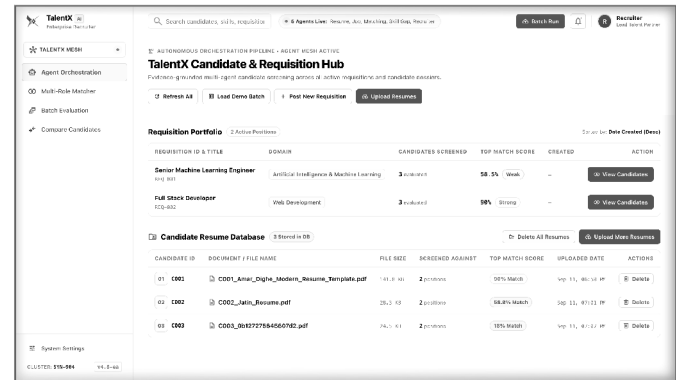


Fig. 3. TalentX Recruiter Dashboard showcasing real-time candidate ranking and match distribution analytics.

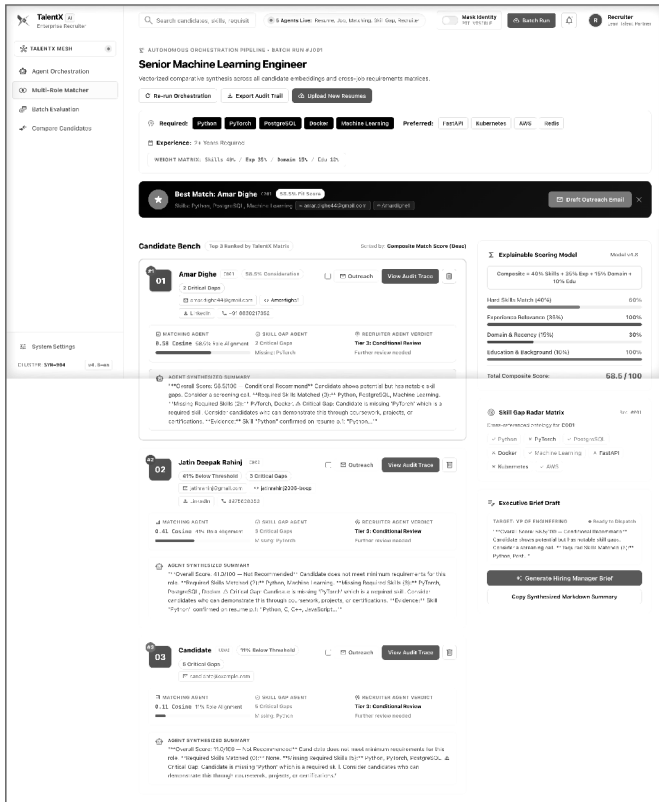


Fig. 4. Candidate Detail Interface highlighting score component breakdowns and evidence text grounding spans.

VII. CONCLUSION & FUTURE WORK

This paper presented **TalentX**, a neuro-symbolic multi-agent AI framework for fair, deterministic, and explainable resume screening. By decoupling LLM semantic extraction from

deterministic score calculation, TalentX eliminates hallucination, guarantees 100% score reproducibility, and enforces demographic fairness through structural PII isolation. Grounding all extractions with verbatim source spans gives recruiters total auditability.

Future work will focus on integrating multi-lingual skill taxonomy mapping and extending privacy-preserving federated candidate matching protocols.

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